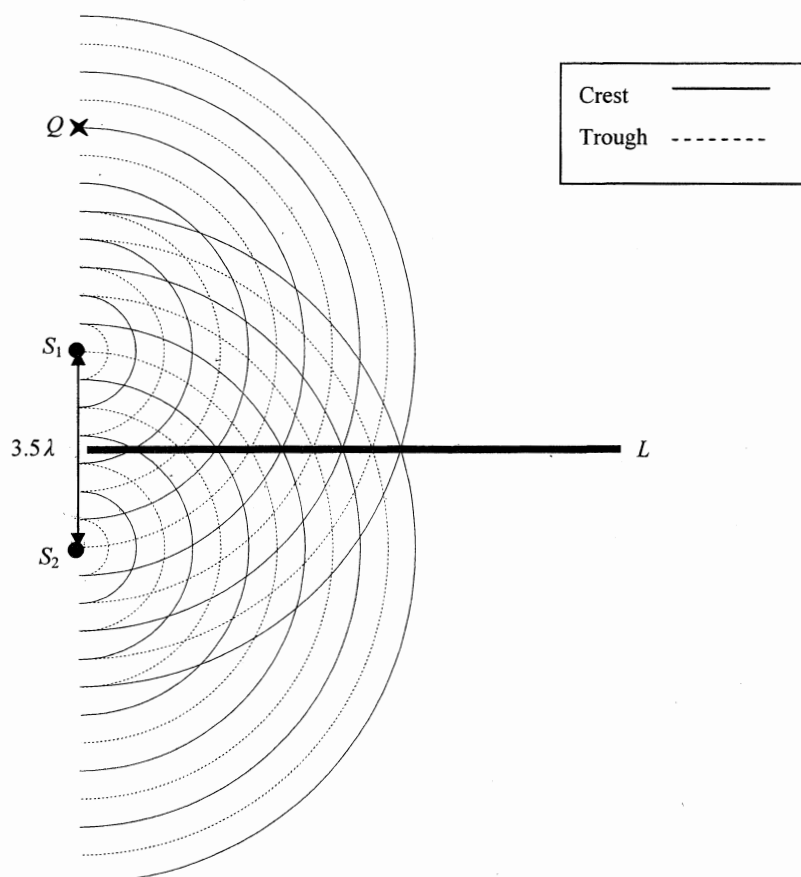


6. In a ripple tank, circular water waves are produced by two vibrators  $S_1$  and  $S_2$  of the same frequency vibrating in phase. Their separation is  $3.5\lambda$ , where  $\lambda$  is the wavelength of the waves. Figure 6.1 shows the two circular waves propagating on the water surface at a certain moment. Line  $L$  is a line connecting all points  $P$  which have path difference  $S_1P - S_2P = 0$ .

Figure 6.1



- (a) Draw and label a line in Figure 6.1 connecting all points  $P$  which have path difference

(i)  $S_1P - S_2P = \lambda$  (label it as  $L_1$ )

(ii)  $S_1P - S_2P = -\frac{3}{2}\lambda$  (label it as  $L_2$ )

What would happen to  $L_1$  and  $L_2$  if the separation between  $S_1$  and  $S_2$  is reduced slightly? (3 marks)

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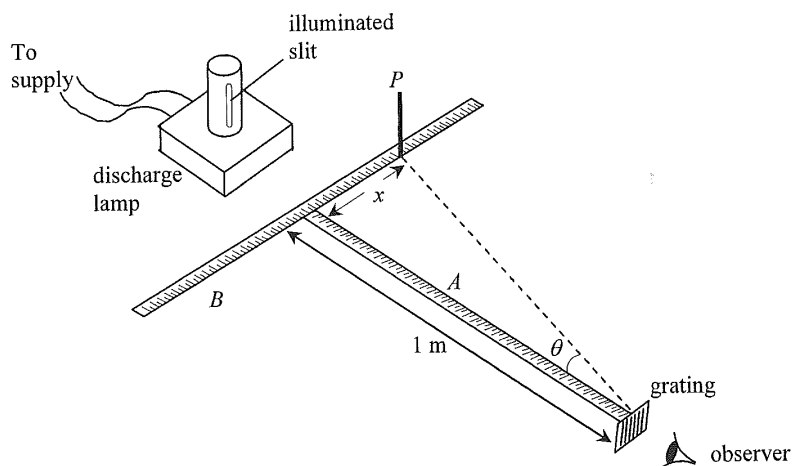
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Answers written in the margins will not be marked.

7. Figure 7.1 shows an experimental set-up to determine the wavelength of monochromatic light emitted from the vertical narrow slit of a discharge lamp.  $A$ ,  $B$  are two mutually perpendicular metre rules on the bench with rule  $A$  pointing towards the lamp. A diffraction grating with vertical lines is placed at one end of rule  $A$ . A vertically mounted pin  $P$  is moved along rule  $B$  until the pin is in line with the diffracted image of the second-order to the observer. The corresponding distance  $x$  is measured for finding the diffraction angle  $\theta$ .

Figure 7.1



The grating has 300 lines per mm and  $x$  is found to be 0.38 m for the **second-order** image.

- (a) (i) Calculate the diffraction angle  $\theta$ . (1 mark)

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- \*(ii) Hence find the wavelength of the light from the lamp. (3 marks)

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- (a) (iii) Give **ONE** advantage of measuring the position of the second-order image instead of the first-order one. (1 mark)

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- (b) In the experiment, the illuminated slit may not be well aligned along metre rule *A*. Suggest one way to reduce this error. (2 marks)

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7. \*(a)

Figure 7.1

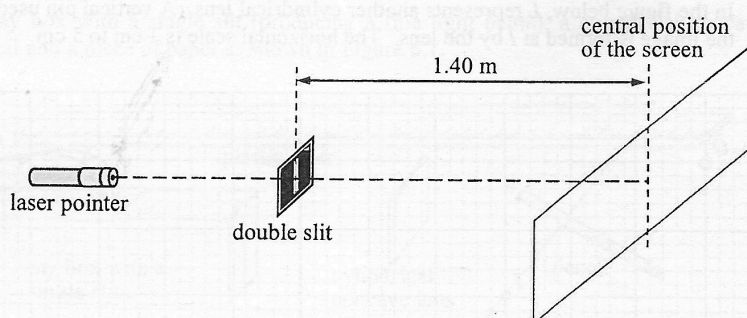


Figure 7.1 shows a set-up for measuring the wavelength  $\lambda$  of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

- (i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)

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The double slit is now replaced by a diffraction grating with 400 lines per mm.

- (ii) Briefly explain why the accuracy of the experiment can be improved. (1 mark)

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- (iii) Only five bright dots are observed on the screen such that the separation between the 1<sup>st</sup> and 5<sup>th</sup> dots is 1.56 m. Find  $\lambda$ . (3 marks)

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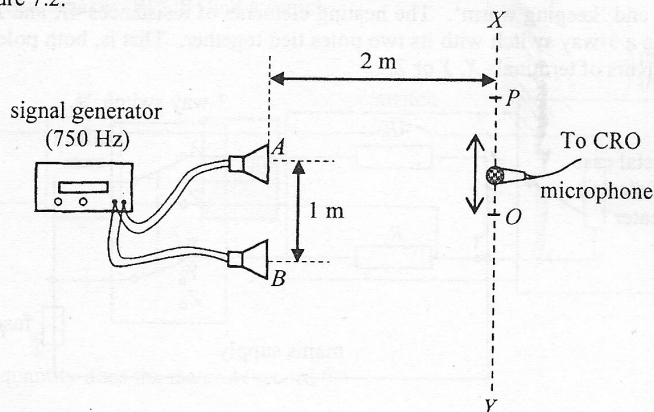
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Answers written in the margins will not be marked.

- (b) To measure the speed of sound in air, a student connects two loudspeakers,  $A$  and  $B$ , to a signal generator as shown in Figure 7.2.

Figure 7.2

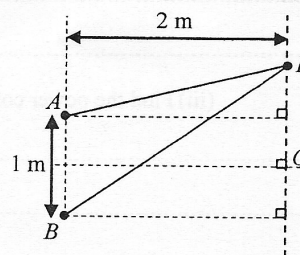


The separation of  $A$  and  $B$  is 1 m. A microphone is used to pick up the sound along the line  $XY$  at a distance of 2 m from the loudspeakers. The central maximum is at point  $O$  while the next maximum is at point  $P$ .

- (i) With reference to the above settings, use the fringe separation equation  $\Delta y = \frac{\lambda D}{a}$  in double-slit interference to find the wavelength  $\lambda$  of sound is not accurate. Explain briefly. (1 mark)

Answers written in the margins will not be marked.

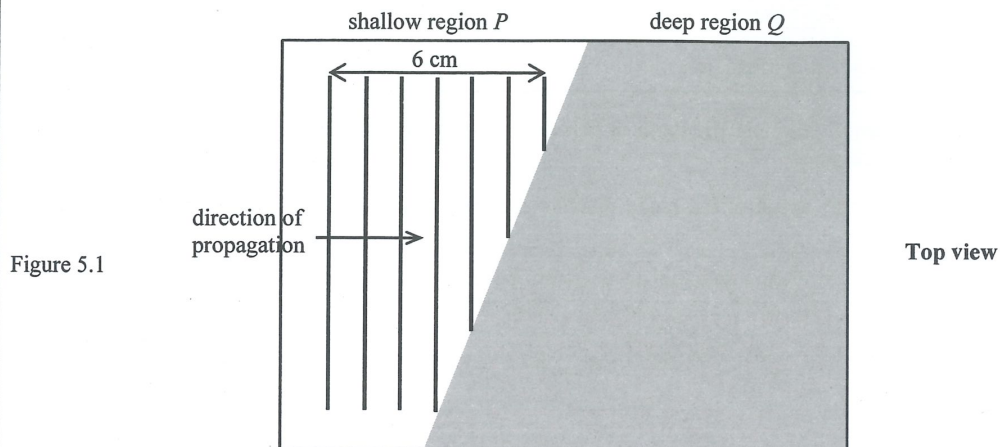
- (ii) The distance between  $O$  and  $P$  is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference  $PB - PA$ , use the results of the experiment to find the speed of sound in air. (3 marks)



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5. A ripple tank has a shallow region  $P$  and a deep region  $Q$ . Straight water wave of frequency 10 Hz is travelling in the shallow region as shown in Figure 5.1 when viewed from above.



- (a) The separation between seven crests in the shallow region is found to be 6 cm as shown.

- (i) Find the wavelength of the wave in the shallow region. (1 mark)

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- (ii) What is the wave speed in the shallow region? (1 mark)

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- (b) The water wave then propagates into the deep region where the wavelength of the wave is double that in the shallow region.

- (i) State the frequency of the water wave in the deep region. (1 mark)

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- (ii) On Figure 5.1, sketch the wave pattern in the deep region. (2 marks)

- (iii) Name the phenomenon occurred across the boundary and explain its cause. (2 marks)

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- \*6. After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation  $d$ .

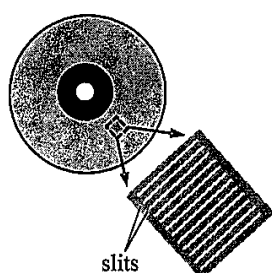


Figure 6.1(a)

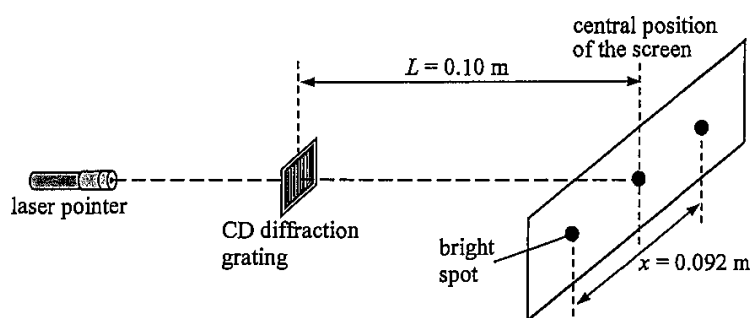


Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen,  $L$ , is 0.10 m, the separation between the first-order maxima,  $x$ , is 0.092 m and the wavelength  $\lambda$  of the red laser light beam used is 650 nm.

(a) Estimate  $d$ .

(3 marks)

(b) State and explain the change in  $x$  if green laser light is used.

(2 marks)

Answers written in the margins will not be marked.